

Don't let our best transportation reporting pass you by



Get weekly updates from Kirsten Korosec and her team about innovation, automation, and more.

SIGN UP NOW

I don't want transportation news.

AI



Tesla signs \$16.5B deal with Samsung to make AI chips

Don't let our best transportation reporting pass you by

Get weekly updates from Kirsten Korosec and her team about innovation, automation, and more.

SIGN UP NOW

I don't want transportation news.

TEC DISRUPT

October 27-29, 2025 | San Francisco

Put your brand in front of 10,000+ tech

s of
park
vation
before

- government's website
- Three weeks after acquiring Windsurf, Cognition offers staff the exit door
 - Roku launches Howdy, a \$2.99 ad-free streaming service
 - Spotify raises subscription prices
 - The uproar over Vogue's AI-generated ad isn't just about fashion

In a later post, Musk said Samsung agreed to allow Tesla to assist in maximizing manufacturing efficiency.

TC

Tech and VC heavyweights join the Disrupt 2025 agenda

Netflix, ElevenLabs, Wayve, Sequoia Capital — just a few of the heavy hitters joining the Disrupt 2025 agenda. They're here to deliver the insights that fuel startup growth and sharpen your edge. Don't miss the 20th anniversary of TechCrunch Disrupt, and a chance to learn from

- Lina Khan points to Figma IPO as vindication of M&A scrutiny
- Your public ChatGPT queries are getting indexed by Google and other search engines

Don't let our best transportation reporting pass you by

Get weekly updates from Kirsten Korosec and her team about innovation, automation, and more.

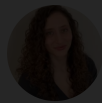
SIGN UP NOW

I don't want transportation news.

Tesla switched over from [Nvidia's Drive platform](#) to its own custom chip in 2019. That chip, known as the [FSD Computer](#) (FSDC) or Hardware 3, launched that year in all of its EVs. The FSDC, made by Samsung, consists of two duplicate systems right next to each other on one board — a design choice meant to create redundancy required for any automated driving system.

Since then, the company's attention to its custom chips has expanded along with its ambitions. The AI chips are the centerpiece of Tesla's push to move from an automaker to an AI and robotics company.

[tesla ai6 chips](#)



Rebecca Bellan

Senior Reporter | [X](#) [🦋](#)

Rebecca Bellan is a senior reporter at TechCrunch, where she covers Tesla and Elon Musk's broader empire,...

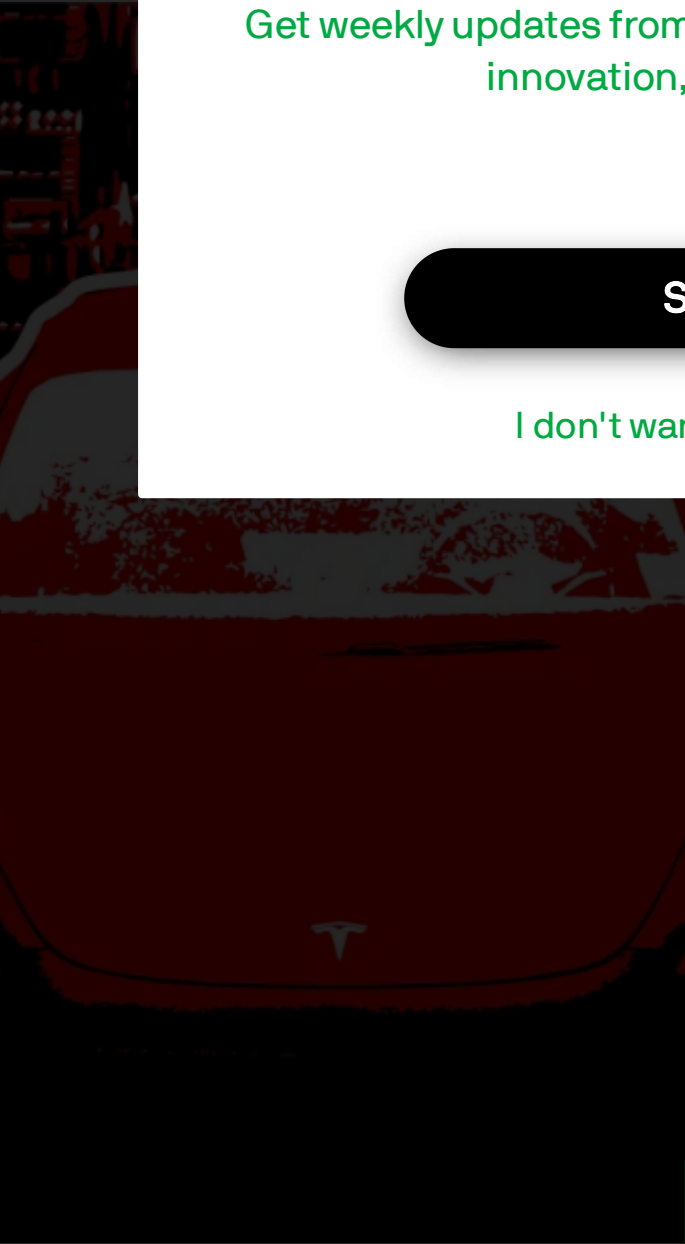
[View Bio](#) >

Don't let our best transportation reporting pass you by

Get weekly updates from Kirsten Korosec and her team about innovation, automation, and more.

SIGN UP NOW

[I don't want transportation news.](#)



Tesla Dojo: Elon Musk's big plan to build an AI supercomputer, explained

For years, Elon Musk has talked about Dojo — the AI supercomputer that will be the cornerstone of Tesla’s AI ambitions. It’s important enough to Musk that in July 2024, he said the company’s AI team would “[double down](#)” on Dojo in the lead-up to [Tesla’s robotaxi reveal](#), which happened in October.

But what exactly is Dojo? And why is it so critical to Tesla’s long-term strategy?

In short: Dojo is Tesla’s custom-built supercomputer that’s designed to train its “Full Self-Driving” neural networks. Beefing up Dojo goes hand-in-hand with Tesla’s goal to reach full self-driving on hundreds of millions of vehicles. It can perform tasks that would otherwise require

Tesla’s now the autonomous vehicle leader in its [2024](#) at the expense of unsupe

Musk’s previous rhetoric has been that Dojo would be the key to achieving Tesla’s goal of full self-driving. Now that Tesla appears close to nearing that goal, Musk has been mum on Dojo.

Instead, ever since August 2024, talk has been around [Cortex](#), Tesla’s “giant new AI training supercluster being built at Tesla HQ in Austin to solve real-world AI.” [Musk has also said](#) it will have “massive storage for video training of FSD & Optimus.”

In [Tesla’s Q4 shareholder deck](#), the company shared updates on Cortex, but nothing on Dojo.

Don't let our best transportation reporting pass you by

Get weekly updates from Kirsten Korosec and her team about innovation, automation, and more.

SIGN UP NOW

I don't want transportation news.

Tech and VC heavyweights join the Disrupt 2025 agenda

Netflix, ElevenLabs, Wayve, Sequoia Capital — just a few of the heavy hitters joining the Disrupt 2025 agenda. They're here to deliver the insights that fuel startup growth and sharpen your edge. Don't miss the 20th anniversary of TechCrunch Disrupt, and a chance to learn from the top voices in tech — grab your ticket now and save up to \$675 before prices rise on August 7.

Advertisement

Newsletters

See More ↗

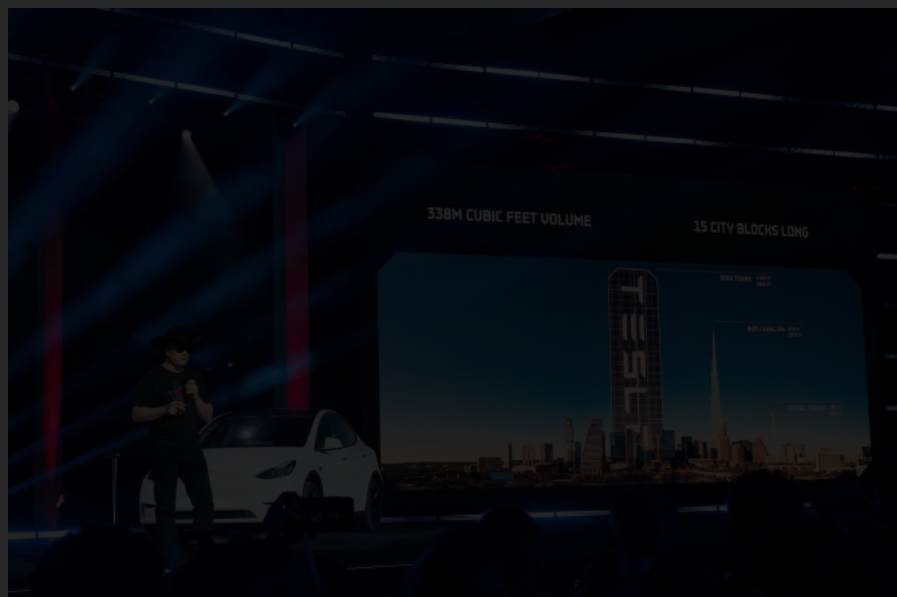
Don't let our best transportation reporting pass you by

Get weekly updates from Kirsten Korosec and her team about innovation, automation, and more.

SIGN UP NOW

I don't want transportation news.

Tesla's Dojo backstory



TechCrunch Mobility

TechCrunch Mobility is your destination for transportation news and insight.

No newsletters selected.

Email address

Subscribe

By submitting your email, you agree to our [Terms](#) and [Privacy Notice](#).

Musk doesn't want Tesla to be just an automaker, or even a purveyor of solar panels and energy storage systems. Instead, he wants Tesla to be an AI company, one that has cracked the code to self-driving cars by mimicking human perception.

Advertisement

Most other companies building autonomous vehicle technology rely on a combination of sensors to perceive the world — like lidar, radar and cameras — as well as high-definition maps to localize the vehicle. Tesla believes it can achieve fully autonomous driving by relying on cameras alone to capture visual data and then use advanced

Don't let our best transportation reporting pass you by

Get weekly updates from Kirsten Korosec and her team about innovation, automation, and more.

SIGN UP NOW

[I don't want transportation news.](#)

As Tesla
at the a
is basic
the gro
2019, bu

Compa
[comme](#)
which t
itself wi

certain conditions — through a more traditional sensor and machine learning approach. Tesla has still yet to produce an autonomous system that doesn't require a human behind the wheel.

About 1.8 million people have paid the hefty [subscription price](#) for Tesla's FSD, which currently costs \$8,000 and has been [priced as high as \\$15,000](#). The pitch is that Dojo-trained AI software will eventually be pushed out to Tesla customers via over-the-air updates. The scale of FSD also means Tesla has been able to rake in millions of miles worth of video footage that it uses to train FSD. The idea there is that the more data Tesla can collect,

the closer the automaker can get to actually achieving full self-driving.

However, some industry experts say there might be a limit to the brute force approach of throwing more data at a model and expecting it to get smarter.

“First of all, there’s an economic constraint, and soon it will just get too expensive to do that,” Anand Raghunathan, Purdue University’s Silicon Valley professor of electrical and computer engineering, told TechCrunch. Further, he said, “Some people claim that we might actually run out of meaningful data to train the models on. More data doesn’t necessarily mean more information, so it depends

on whether you’re trying to create a model that is able to handle a better model.

Raghunathan said of more data, “At least in the power market, Tesla’s supercomputer

Don't let our best transportation reporting pass you by

Get weekly updates from Kirsten Korosec and her team about innovation, automation, and more.

SIGN UP NOW

[I don't want transportation news.](#)

What is a supercomputer?

Dojo is Tesla’s supercomputer system that’s designed to function as a training ground for AI, specifically FSD. The name is a nod to the space where martial arts are practiced.

A supercomputer is made up of thousands of smaller computers called nodes. Each of those nodes has its own CPU (central processing unit) and GPU (graphics processing unit). The former handles overall management of the node, and the latter does the complex stuff, like splitting tasks into multiple parts and working on them simultaneously.

GPUs are essential for machine learning operations like those that power FSD training in simulation. They also power large language models, which is why the rise of generative AI has made Nvidia the most valuable company on the planet.

Even Tesla buys Nvidia GPUs to train its AI (more on that later).

Why does Tesla need a supercomputer?

Tesla's vision-only approach is the main reason

Tesla needs to be behind the data to vehicle means have to continu velocity

In other duplica function

To get the video data collected from its cars around the world and run millions of simulations to train its model on the data.

Dojo pics pic.twitter.com/Lu8YiZXo8c

— Elon Musk (@elonmusk) [July 23, 2024](#)

Tesla appears to rely on Nvidia to power its current Dojo training computer, but it doesn't want to have all its eggs in one basket — not least because Nvidia chips are expensive. Tesla also hopes to make something better that increases bandwidth and decreases latencies. That's why the

Don't let our best transportation reporting pass you by

Get weekly updates from Kirsten Korosec and her team about innovation, automation, and more.

SIGN UP NOW

[I don't want transportation news.](#)

automaker's AI division decided to come up with its own custom hardware program that aims to train AI models more efficiently than traditional systems.

At that program's core is Tesla's proprietary D1 chips, which the company says are optimized for AI workloads.

Tell me more about these chips



Don't let our best transportation reporting pass you by

Get weekly updates from Kirsten Korosec and her team about innovation, automation, and more.

SIGN UP NOW

I don't want transportation news.

Tesla unveiled its D1 chip, a silicon square the size of a palm, on AI Day in 2021. The D1 chip entered into production as of at least May this year. The Taiwan Semiconductor Manufacturing Company (TSMC) is manufacturing the chips using 7 nanometer semiconductor nodes. The D1 has 50 billion transistors and a large die size of 645 millimeters squared, according to Tesla. This is all to say that the D1 promises to be extremely powerful and efficient and to handle complex tasks quickly.

“We can do compute and data transfers simultaneously, and our custom ISA, which is the instruction set architecture, is fully optimized for machine learning workloads,” said Ganesh Venkataramanan, former senior director of Autopilot hardware, at Tesla’s 2021 AI Day. “This is a pure machine learning.”

The D1 is still not as powerful as Nvidia’s A100 chip, though, which is also manufactured by TSMC using a 7 nanometer process. The A100 contains 54 billion transistors and has a die size of 826 square millimeters, so it performs slightly better than Tesla’s D1.

To get a power, Tesla into one system petaflop bandwidth necess can third made u make u cabinet 2022, Te multiple supercomputer.

Don't let our best transportation reporting pass you by

Get weekly updates from Kirsten Korosec and her team about innovation, automation, and more.

SIGN UP NOW

[I don't want transportation news.](#)

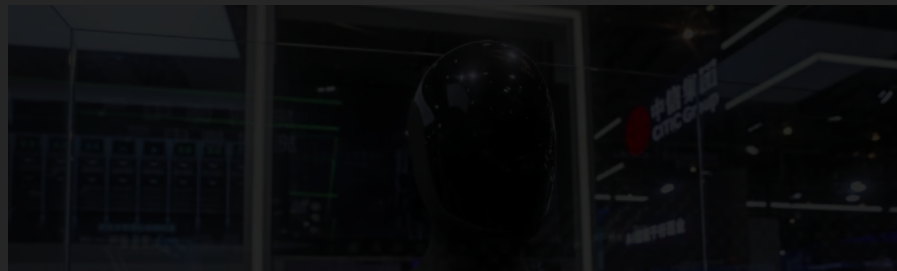
Tesla is also working on a next-gen D2 chip that aims to solve information flow bottlenecks. Instead of connecting the individual chips, the D2 would put the entire Dojo tile onto a single wafer of silicon.

Tesla hasn’t confirmed how many D1 chips it has ordered or expects to receive. The company also hasn’t provided a timeline for how long it will take to get Dojo supercomputers running on D1 chips.

In response to [a June post on X](#) that said: “Elon is building a giant GPU cooler in Texas,” Musk replied

that Tesla was aiming for “half Tesla AI hardware, half Nvidia/other” over the next 18 months or so. The “other” could be AMD chips, per [Musk’s comment in January](#).

What does Dojo mean for Tesla?



Don't let our best transportation reporting pass you by

Get weekly updates from Kirsten Korosec and her team about innovation, automation, and more.

SIGN UP NOW

[I don't want transportation news.](#)

Startups

AI

Venture

Apps

Apple

Events

Podcasts

Newsletters



that it’s often difficult to get the GPUs.” He said he

was “quite concerned about actually being able to get steady GPUs when we want them, and I think this therefore requires that we put a lot more effort on Dojo in order to ensure that we’ve got the training capability that we need.”

That said, Tesla is still buying Nvidia chips today to train its AI. In June, [Musk posted on X](#):

Of the roughly \$10B in AI-related expenditures I said Tesla would make this year, about half is internal, primarily the Tesla-designed AI inference computer and sensors present in all of our cars, plus Dojo. For building the AI training superclusters, Nvidia hardware is about 2/3 of the cost. We’re also looking at other options for purchase.

“Inference performance is a key metric for the AI inference stack, and we’re looking for.”

Dojo is a several billion dollar success story.

In the long term, Tesla’s new business model based on its AI division. Musk has said that the first version of Dojo will be tailored for Tesla computer vision labeling and training, which is great for FSD and for training [Optimus](#), Tesla’s humanoid robot. But it wouldn’t be useful for much else.

[Musk has said](#) that future versions of Dojo will be more tailored to general-purpose AI training. One potential problem with that is almost all AI software out there has been written to work with GPUs. Using Dojo to train general-purpose AI models would require rewriting the software.

Don't let our best transportation reporting pass you by

Get weekly updates from Kirsten Korosec and her team about innovation, automation, and more.

SIGN UP NOW

[I don't want transportation news.](#)

That is, unless Tesla rents out its compute, similar to how AWS and Azure rent out cloud computing capabilities. Musk also noted during Q2 earnings that he sees “a path to being competitive with Nvidia with Dojo.”

A September 2023 report from Morgan Stanley predicted that Dojo could add \$500 billion to Tesla’s market value by unlocking new revenue streams in the form of robotaxis and software services.

★ SPONSORED



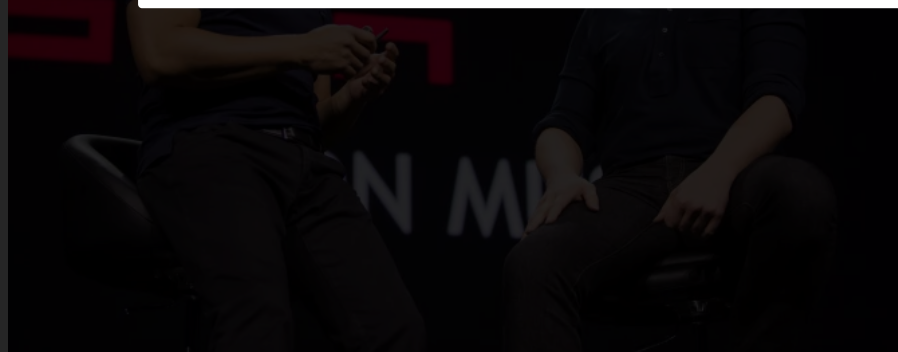
Design agents are here — Lovart launches the first end-to-end AI canvas for design

Don't let our best transportation reporting pass you by

Get weekly updates from Kirsten Korosec and her team about innovation, automation, and more.

SIGN UP NOW

I don't want transportation news.



NVIDIA CEO JENSEN HUANG AND TESLA CEO ELON MUSK AT THE GPU TECHNOLOGY CONFERENCE IN SAN JOSE, CALIFORNIA.

IMAGE CREDITS:KIM KULISH/CORBIS VIA GETTY IMAGES / GETTY IMAGES

Reuters reported last year that Tesla began production on Dojo in July 2023, but a June 2023

[post](#) from Musk suggested that Dojo had been “online and running useful tasks for a few months.”

Around the same time, Tesla said it expected Dojo to be one of the top five most powerful supercomputers by February 2024 — a feat that has yet to be publicly disclosed, leaving us doubtful that it has occurred.

The company also said it expects Dojo’s total compute to reach 100 exaflops in October 2024. (One exaflops is equal to 1 quintillion computer operations per second. To reach 100 exaflops, and assuming that one D1 can achieve 362 teraflops, Tesla would need more than 276,000 D1s, or a

Tesla also [million](#) gigafactory

In May 2023, Tesla’s “supercomputer” cluster, Dojo, the

Just after Musk [posted on X](#) that the automaker’s AI team is using Tesla HW4 AI computer (renamed AI4), which is the hardware that lives on Tesla vehicles, in the training loop with Nvidia GPUs. He noted that the breakdown is roughly 90,000 Nvidia H100s plus 40,000 AI4 computers.

“And Dojo 1 will have roughly 8k H100-equivalent of training online by end of year,” he continued. “Not massive, but not trivial either.”

Don't let our best transportation reporting pass you by

Get weekly updates from Kirsten Korosec and her team about innovation, automation, and more.

SIGN UP NOW

[I don't want transportation news.](#)



Design agents are here — Lovart launches the first end-to-end AI canvas for design

Sponsored by Lovart

Tesla hasn't provided updates as to whether it has gotten those chips online and running Dojo. During the company's fourth-quarter 2024 earnings call, no one mentioned Dojo. However, Tesla said it completed the deployment of Cortex in Q4 and that it was Cortex that helped enable V13 of supervised FSD.

This story originally published August 3, 2024, and we will update it as more information becomes available.

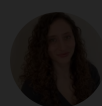
Topics:

Don't let our best transportation reporting pass you by

Get weekly updates from Kirsten Korosec and her team about innovation, automation, and more.

SIGN UP NOW

I don't want transportation news.



Rebecca Bellan is a senior reporter at TechCrunch, where she covers Tesla and Elon Musk's broader empire,...

[View Bio](#) >

KEEP READING

Our top 4 Tesla AI day predictions

Don't let our best transportation reporting pass you by

Get weekly updates from Kirsten Korosec and her team about innovation, automation, and more.

SIGN UP NOW

I don't want transportation news.

AI Day is a key event to watch for capabilities across different business segments, with the ultimate goal of recruiting the brightest minds and engineers.

Even without Tesla's star head of AI Andrej Karpathy, who [left the company in July](#), expect this event to be full of technical jargon and big promises from Tesla's executives, including the big dog himself, CEO Elon Musk.

We'll talk about what we're expecting to hear about based on recent news, as well as [last year's AI day event](#). But first, how to watch.

killing website traffic

Sarah Perez - 12 hours ago

AI



First impressions of Alexa+, Amazon's upgraded, AI-powered digital assistant

Sarah Perez - 13 hours ago

AI



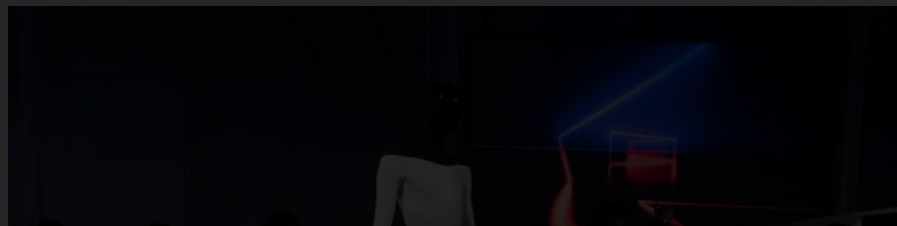
Google takes on ChatGPT's Study Mode with new 'Guided Learning' tool in Gemini

Aisha Malik - 14 hours ago

Tesla's AI Day is scheduled for September 30 and should begin around 5 p.m. PT, although Tesla's events are often notorious for running late. Remote viewers can watch it livestreamed on [Tesla's YouTube channel](#) or on its website.

Now, onto the forecasts.

An Optimus prototype



THE TESLA
MANUFACTU
OPENING F
CREDITS:
IMAGE CRE

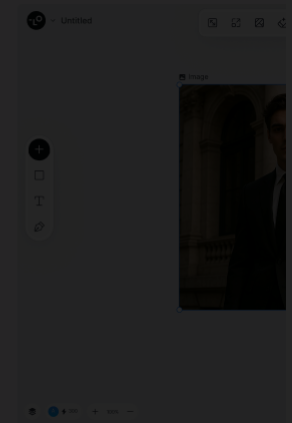
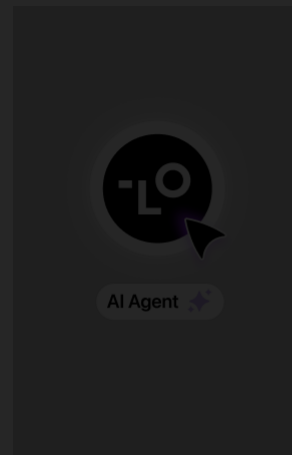
At last y
[product](#)

that is “intended to be friendly and navigate through a world built for humans,” according to Musk.

Tesla AI Day was originally scheduled for August 19, but [Musk pushed the date back](#) in the hopes of having a working Optimus prototype by this week. On September 20, [Musk reiterated his goal](#) to have the prototype ready by the end of the month.

During Tesla's first-quarter earnings call this year, the celebrity executive told investors the company continues to work on the 5'8" robot, which will rely on Tesla's work in neural networks and its Dojo advanced supercomputer (more on that in a bit), to

Sponsored by Lovart



★ SPONSORED

Design agents are here — Lovart launches the first end-to-end AI canvas for design

Don't let our best transportation reporting pass you by

Get weekly updates from Kirsten Korosec and her team about innovation, automation, and more.

SIGN UP NOW

I don't want transportation news.

handle the drudgery of human existence, AKA everyday tasks like grocery shopping. At the time, Musk said Optimus would be “worth more than the car business, worth more than FSD.”

FSD, or Full Self-Driving, is Tesla’s advanced driver assistance system (ADAS) that is definitely not fully autonomous, but can perform some autonomous driving tasks for drivers [willing to shell out \\$15,000](#). Musk has also said that Tesla, without FSD, is worth [“basically zero.”](#)

During [Tesla’s Austin gigafactory opening](#) in April, Musk said Optimus, along with a new wave of products, will be introduced in 2023. Which brings us to the

If it mar
Optimu
about 5
be wea
the bot

Unless
progres
like tas
product
does, w

Don't let our best transportation reporting pass you by

Get weekly updates from Kirsten Korosec and her team about innovation, automation, and more.

SIGN UP NOW

[I don't want transportation news.](#)

TC

Tech and VC heavyweights join the Disrupt 2025 agenda

Netflix, ElevenLabs, Wayve, Sequoia Capital — just a few of the heavy hitters joining the Disrupt 2025 agenda. They’re here to deliver the insights that fuel startup growth and sharpen your edge. Don’t miss the 20th anniversary of TechCrunch Disrupt, and a chance to learn from the top voices in tech — grab your

ticket now and save up to \$675 before prices rise on August 7.

San Francisco | October 27-29, 2025

REGISTER NOW

Musk has been promising to solve full self-driving for years now — in 2016, he said Tesla was less than two years away from solving “complete autonomy.” FSD still does not have full self-driving capabilities, and those shortcomings are causing problems for Tesla. The automaker is facing state and federal scrutiny for Autopilot, its lesser ADAS version, and

FSD, even as multiple advertisers

Update

Don't let our best transportation reporting pass you by

Get weekly updates from Kirsten Korosec and her team about innovation, automation, and more.

SIGN UP NOW

I don't want transportation news.

WHAT TESLA'S AUTOPILOT SEES. IMAGE CREDITS: TESLA

IMAGE CREDITS: TESLA

Musk also recently tweeted that Tesla's AI team has an end-of-month deadline for “actually smart” summon and auto park, both of which are features of Tesla's Enhanced Autopilot, its \$6,000 ADAS. Smart summon, which Musk's tweet implies has been operating at a less than perfect performance, promises to navigate the car in complex

environments and parking spaces, “maneuvering around objects as necessary to come find you in a parking lot.” Autopark helps “automatically parallel or perpendicular park your car, with a single touch,” [according to Tesla’s website](#).

Tesla is likely to [reiterate its goal to get to a widespread rollout of FSD beta](#) in North America by the end of this year. We’re expecting the automaker to also provide specific updates to the next version of FSD beta, like having the vehicle take note of surrounding cars and [increase speed to keep up with traffic](#).

A Tes

During T
[opening](#)
said 20
2023 w
product

At the ti
that wo
later, du
more al
purpose

built without a steering wheel or pedals. It will also offer riders the lowest cost per mile of any other transport, and be cheaper than a subsidized bus ticket, the CEO promised.

Tesla has been talking about a robotaxi concept since 2016, when [Musk came out with his Master Plan Part Deux](#). The CEO imagined a Tesla Network taxi service that would allow Tesla owners to throw their cars in the autonomous ring to generate income on their behalf while they’re at work or on vacation.

Don't let our best transportation reporting pass you by

Get weekly updates from Kirsten Korosec and her team about innovation, automation, and more.

SIGN UP NOW

[I don't want transportation news.](#)

Will Musk finally provide specifics about what these robotaxis will actually look like, how they will be commercialized, or whether Tesla has set a date for production? Find out on Friday.

Updates to the Dojo supercomputer

Don't let our best transportation reporting pass you by

Get weekly updates from Kirsten Korosec and her team about innovation, automation, and more.

SIGN UP NOW

[I don't want transportation news.](#)

SCREENSHOTS
FOR DOJO
IMAGE CRE

This is v

Last year
D1, that
superco

dependent on Dojo, the neural network training supercomputer that Musk says can process vast amounts of camera imaging data four times faster than other computing systems. Having a super fast computer is important for Tesla because it relies solely on cameras, rather than lidar and radar, for autonomous operations. This speed depends on the processing power of D1.

In late August, Tesla participated in Hot Chips 34, a conference on semiconductors, during which the company revealed new information about Dojo's AI system architecture and supercomputer. Serve the Home, [an industry blog](#), published Tesla's slides,

which detail specifics about how Dojo's distributed system facilitates machine learning and how this will lead Tesla to generalized autonomy. The presentations were highly technical, so we'll skip the specifics on hardware and capabilities, but we're expecting much of the same information to be shared on AI Day, as well as how it all contributes to performance.

Surprise announcements

Again, Musk has repeatedly hinted at a wave of new products coming forth next year, so we wouldn't be surprised if he came out with something truly wacky and

We'll be check b news is

Topics:

Don't let our best transportation reporting pass you by

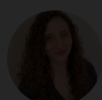
Get weekly updates from Kirsten Korosec and her team about innovation, automation, and more.

SIGN UP NOW

[I don't want transportation news.](#)

Advertisement

[f](#) [X](#) [in](#) [📺](#) [✉](#) [🔗](#)



Rebecca Bellan

Senior Reporter | [X](#) [🦋](#)

Rebecca Bellan is a senior reporter at TechCrunch, where she covers Tesla and Elon Musk's broader empire,...

View Bio >

Loading the next article



TechCrunch

Staff

Contact Us

Advertise

Terms of Service

Privacy Policy

RSS Terms of

OpenAI

Perplexity

Cisco

DeepMind

Windsurf

ayoffs

PT



Don't let our best transportation reporting pass you by

Get weekly updates from Kirsten Korosec and her team about innovation, automation, and more.

SIGN UP NOW

I don't want transportation news.